Global Procurement and Materials Department
Installation Guide
1.0

OpenFOAM Setup Guide

Global Procurement and Materials Department – MPS Team

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KEYSIGHT TECHNOLOGIES

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Change History

Revision Changes			
Rev. #	**Last Updated**	**Author**	**Comments**
1.0	27 April, 2026	‘Affan Najiy Bin Rusdi	Initial Copy

1.0 Introduction

This document provides a step-by-step guide to installing Windows Subsystem for Linux (WSL), Ubuntu, and OpenFOAM, along with the mesh utility scripts and aliases for daily use.

Compatible with ESI OpenFOAM v2506, Ubuntu 24.04 LTS, Windows 10/11

Quick-Start Checklist

Use this checklist each time you start a new work session:

- Open Windows Terminal and select Ubuntu (or run: wsl -d Ubuntu).
- Source the OpenFOAM environment: of2506
- Navigate to the project directory: myDir
- Change into your specific case directory: cd <case_name>
- Launch the GUI: openfoamUI — or run scripts directly.

Note: For persistent alias changes, always edit ~/.bash_aliases and then run source ~/.bash_aliases.

2.0 Windows Subsystem for Linux (WSL)

2.1 Installing WSL

Windows Subsystem for Linux (WSL) allows you to run a Linux environment directly on Windows without needing a virtual machine.

1. Open PowerShell as Administrator.
2. Run the following command:

```
wsl --install
```

3. Restart your device when prompted

2.2 Installing Ubuntu Linux Distribution

After WSL is installed, set up an Ubuntu distribution:

1. Open PowerShell as Administrator.
2. Run the following command:

```
wsl --install Ubuntu
```

3. Restart your device again once installation completes.

4. When Ubuntu first launches, you will be prompted to create a Unix username and password. Choose a memorable username, it does not need to match your Windows credentials.

2.3 Opening WSL (Ubuntu Terminal)

There are two ways to open your Ubuntu terminal:

Option A: Via the Windows Terminal UI

1. Open Windows Terminal.
2. Click the down arrow (v) next to the tab bar.
3. Select Ubuntu from the list.

Option B: Via Command

1. Open Windows Terminal or PowerShell.
2. Run:

```
wsl -d Ubuntu
```

3.0 Installing OpenFOAM

All commands in this section must be run inside the Ubuntu terminal. Run them in the exact order shown.

Step 1: Add the OpenFOAM Debian repository

```
curl -s https://dl.openfoam.com/add-debian-repo.sh | sudo bash
```

Step 2: Update the package list

```
sudo apt-get update
sudo apt-get upgrade
```

Step 3: Install OpenFOAM

```
sudo apt-get install openfoam2506-default
```

Note: The installation may take several minutes depending on your internet connection. Accept any prompts with Y when asked.

4.0 Placing the Project Files

Create the OpenFOAM utilities directory and place the project scripts inside it.

Step 1: Navigate to the root directory

```
cd /
```

Step 2: Create the directory

```
mkdir -p /mnt/c/OpenFOAM
mkdir 01_utilities
```

Step 3: Place the scripts

Clone or manually copy the repository files so the directory contains at minimum:

- 01_utilities/generateBackgroundMesh.py
- 01_utilities/generateSnappyHexMeshDict.py
- 01_utilities/openfoam_ui.py

Note: The path /mnt/c/OpenFOAM corresponds to C:\OpenFOAM on your Windows file system.

5.0 Installing Python Dependencies

The mesh utility scripts require Python 3 along with several system packages. Run the following inside the Ubuntu terminal:

```
sudo apt-get install python3-pip
sudo apt-get install python3-numpy python3-tk
```

To install any additional Python library in future:

```
sudo apt-get install python3-<library_name>
```

Note: Replace <library_name> with the actual package name, for example python3-scipy.

6.0 Creating Bash Aliases

Aliases let you run long commands with short, memorable names. Follow these steps inside the Ubuntu terminal.

Step 1: Open the aliases file in vi

```
vi ~/.bash_aliases
```

Step 2: Enter Insert mode

Press the i key to enter Insert mode. You will see -- INSERT -- at the bottom of the terminal.

Step 3: Add the alias definitions

Type or paste the following content:

```
alias myDir="cd /mnt/c/OpenFOAM"
# OpenFOAM version
alias of2506="source /usr/lib/openfoam/openfoam2506/etc/bashrc"

# Python scripts for OpenFOAM workflow
alias generateBackgroundMesh="python3
/mnt/c/OpenFOAM/01_utilities/generateBackgroundMesh.py"
alias generateSnappyHexMeshDict="python3
/mnt/c/OpenFOAM/01_utilities/generateSnappyHexMeshDict.py"
alias openfoamUI="python3 /mnt/c/OpenFOAM/01_utilities/openfoam_ui.py"
```

Step 4: Save and exit

1. Press Esc to exit Insert mode.
2. Type

```
:wq
```

and press Enter to save and quit.

Quick vi command reference:

Command	Action
:q	Quit without saving
:q!	Force quit and discard changes
:x	Save and quit (only if changes were made)
i	Enter Insert mode
Esc	Exit Insert mode

Step 5: Apply the aliases to your current session

```
source ~/.bash_aliases
```

Note: You only need to run source once per terminal session. New terminal sessions will load the aliases automatically.

7.0 Launching OpenFOAM

Every time you start a new Ubuntu terminal session, follow these two steps before running any OpenFOAM command or utility:

Step 1: Source the OpenFOAM environment

```
of2506
```

Step 2: Navigate to the working directory

```
myDir
```

Note: You must source of2506 in every new terminal session. OpenFOAM commands such as blockMesh and snappyHexMesh will not be available until you do.

8.0 Running the Mesh Utilities

Once the OpenFOAM environment is sourced and you are in the working directory, you can run the utilities using the aliases defined in Section 7.

8.1 Graphical User Interface (Recommended)

The GUI wraps both utilities in a single window with form inputs and a live output log.

```
openfoamUI
```

The GUI provides two tabs:

- Background Mesh — generates the blockMeshDict and runs blockMesh based on an STL file and grid sizes.
- SnappyHexMesh Dict — guides you through generating a snappyHexMeshDict interactively.

8.2 Command-Line Scripts

Both utilities can also be run directly from the terminal with arguments.

Generate Background Mesh

Navigate to your OpenFOAM case directory first, then run:

```
cd /path/to/your/case
generateBackgroundMesh -stlPath /path/to/geometry.stl -dx 0.05 -dy 0.05 -dz 0.05
```

- Arguments:
- -stlPath Path to the STL geometry file.
- -dx Base grid size in the X direction (positive number).
- -dy Base grid size in the Y direction (positive number).
- -dz Base grid size in the Z direction (positive number).

Generate snappyHexMeshDict

Navigate to your OpenFOAM case directory first, then run:

```
cd /path/to/your/case
generateSnappyHexMeshDict
```

The script will prompt you interactively for geometry files, refinement levels, snap settings, and layer addition options.

9.0 Troubleshooting

Symptom	Resolution
OpenFOAM commands not found (e.g. blockMesh: command not found)	You have not sourced the environment. Run: of2506
wsl --install fails or WSL version is 1	Enable the 'Virtual Machine Platform' Windows feature and update WSL: wsl --update
python3-tk not found / GUI does not open	Run: sudo apt-get install python3-tk then retry openfoamUI
surfaceCheck: command not found	Source of2506 before running any mesh utility.
Permission denied when running scripts	Ensure the scripts are readable: chmod +r /mnt/c/OpenFOAM/01_utilities/*.py

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